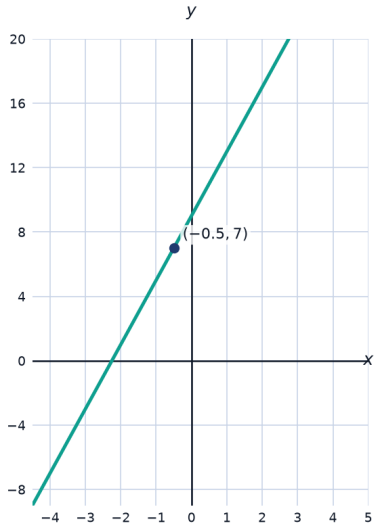
Trigonometric Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

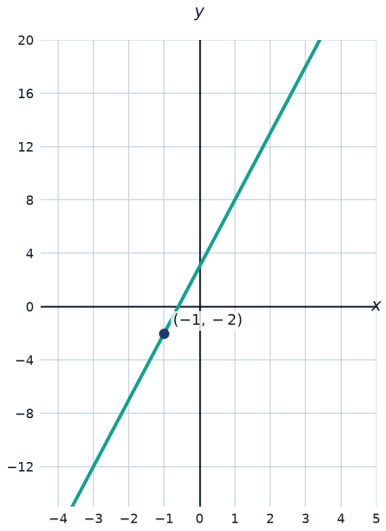
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(2)$.



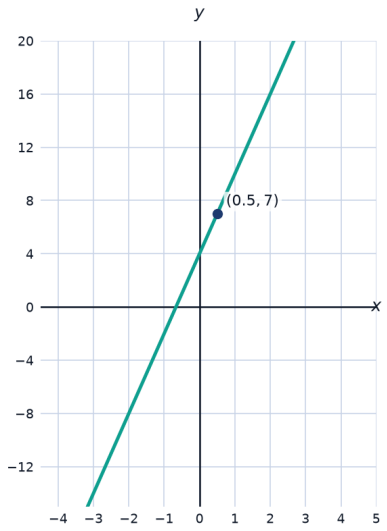
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for key features of secant, cosecant, and cotangent graphs consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(2)$.

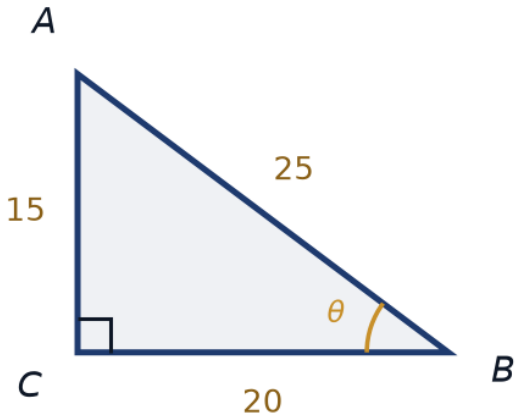


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for graphing secant, cosecant, and cotangent functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The cosine function is symmetric about the y-axis because it is:.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Unit 3 · Trigonometric and Polar Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

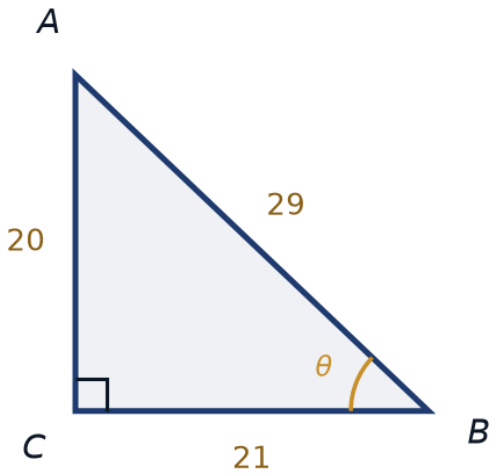
Prove the trigonometric identity $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$ for all values of θ where both sides are defined.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the trigonometric equation $2\cos^2 x - \sin x - 1 = 0$ for all x in the interval $[0, 2\pi)$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Prove the trigonometric identity $\frac{\sin(2x)}{1+\cos(2x)} = \tan x$ and state the domain exclusions.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{3x+4}{2x-5}$, and state the domain of f^{-1}.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $f(x) = \frac{2}{x-1}$ and $g(x) = \sqrt{x+5}$, find (a) the value of $(f \circ g)(4)$, and (b) the domain of $g(x)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

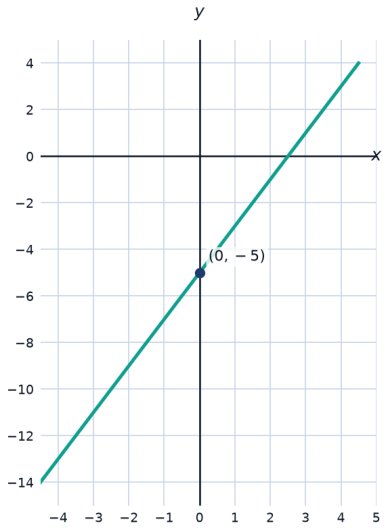
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Polynomial Functions

Determine whether the function $g(x) = \frac{x^3}{x^2+4}$ is even, odd, or neither.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Rational Expressions

If $f(x) = 2x - 5$ and $g(x) = x^2 + 3$, find the expression for $(f \circ g)(x)$ and solve for all x such that $(f \circ g)(x) = 25$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find all solutions to $2x^2 - 6x + 5 = 0$ using the quadratic formula, and determine whether the roots are real or complex.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

[Functions & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Conic Sections

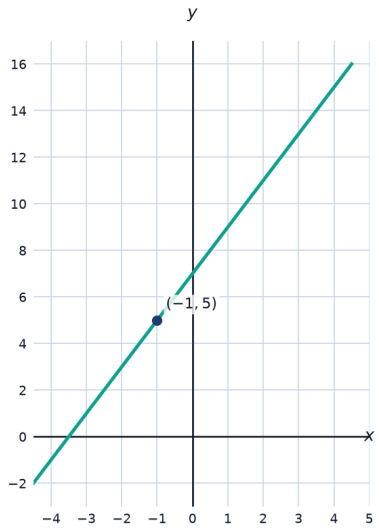
[Quadratic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(2)$.



a. 11

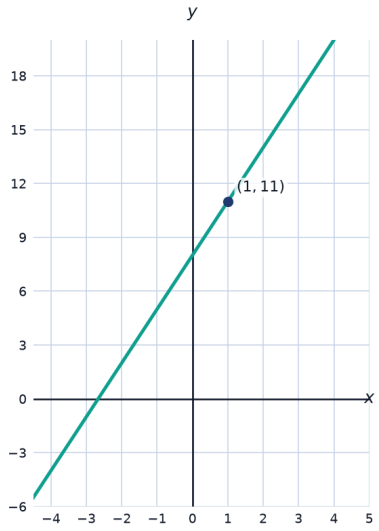
b. 13

c. 4

d. 15

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 8$ in graphing sine and cosine functions, evaluate $f(2)$.



a. 14

b. 17

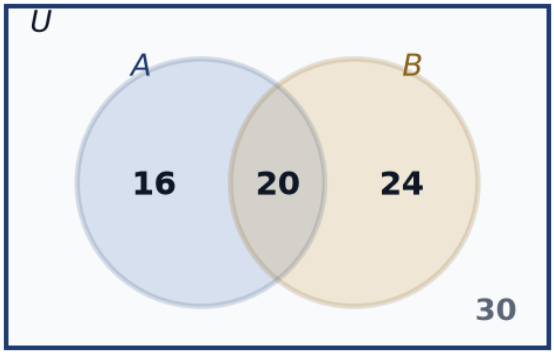
c. 6

d. 18

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for key features of sine, cosine, and tangent graphs consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?



a. 20.5

b. 21.5

c. 22.5

d. 24.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for graphing sine and cosine functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for graphing tangent functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the maximum value of the function $f(x) = \sqrt{3}\sin(2x) - \cos(2x) - 2\sin x + 2\sqrt{3}\cos x$ for all real numbers x?

a. 3

b. 2

c. 6

d. 1

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = 4\cos^2(x) + b\sin(x) + c$ where b and c are real constants with $b > 0$. If the maximum value of $f(x)$ is 10 and the minimum value of $f(x)$ is 1, what is the value of c?

a. 5

b. $\frac{11}{2}$

c. 9

d. 4

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The function $f(x) = a\cos(bx) + d$, where $a, b,$ and d are positive constants, has a period of π, a maximum value of 7, and satisfies $f\left(\frac{\pi}{6}\right) = 4$. What is the value of $f\left(\frac{\pi}{3}\right)$?

a. -2

b. 1

c. 4

d. -5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a right-angled triangle, one of the acute angles is 30°. What is the measure of the other acute angle?

a. 30°

b. 45°

c. 60°

d. 90°

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State and prove De Moivre's theorem: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

a. By induction: base trivial; step uses $(\cos\theta + i\sin\theta)^{n+1} = (\cos n\theta + i\sin n\theta)(\cos\theta + i\sin\theta) = \cos(n+1)\theta + i\sin(n+1)\theta$. ✓

b. Not provable by induction

c. Only for $n \in \mathbb{Z}^+$

d. Use Euler's formula directly

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Prove the product formula: $\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$.

a. Only if $A = B$

b. Use integration

c. Not true

d. From $\cos(A-B) = \cos A\cos B + \sin A\sin B$ and $\cos(A+B) = \cos A\cos B - \sin A\sin B$. Add and halve. ✓

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Prove the sum-to-product formula: $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$.

a. Not true

b. Only for $A = B$

c. Set $A = M+N, B = M-N$; then $\sin(M+N)+\sin(M-N) = 2\sin M\cos N$. Substitute $M = (A+B)/2, N = (A-B)/2$. ✓

d. Use calculus

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Prove the identity $\sin 3A = 3\sin A - 4\sin^3 A$ using the angle addition formula.

a. Not an identity

b. Only for $A < \pi/2$

c. Use calculus

d. $\sin(2A+A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1-2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find all values of θ such that $\sin\theta, \cos\theta, \tan\theta$ are in arithmetic progression.

a. $\theta \approx 0.6435$ rad (a special angle satisfying $2\cos\theta = \sin\theta + \tan\theta$)

b. $\theta = \pi/3$

c. No solution exists

d. $\theta = \pi/4$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the half-angle formula for $\cos(\theta/2)$.

a. $\pm\sqrt{\frac{1+\cos\theta}{2}}$

b. $\cos\theta/2$

c. $\pm\sqrt{\frac{1-\cos\theta}{2}}$

d. $\sqrt{1+\cos\theta}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the sum-to-product formula for $\cos A - \cos B$.

a. $2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$

b. $2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$

c. $2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$

d. $-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Using the product-to-sum formula, simplify $\sin A\cos B$.

a. $\cos(A-B)$

b. $\frac{1}{2}[\sin(A+B)+\sin(A-B)]$

c. $\sin(A+B)$

d. $\frac{1}{2}[\cos(A-B)-\cos(A+B)]$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Evaluate $\sin 3\theta$ in terms of $\sin\theta$.

a. $3\sin\theta - 4\sin^3\theta$

b. $4\sin^3\theta - 3\sin\theta$

c. $3\sin\theta + 4\sin^3\theta$

d. $3\sin\theta\cos^2\theta$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

Find the inverse of $y = \cos x$ on $[0, \pi]$.

a. $x = \arcsin y$

b. $x = \pi - y$

c. $x = 2\pi - \arccos y$

d. $x = \arccos y$ for $y \in [-1, 1]$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Conic Sections

Find the inverse function: if $y = \sin x$, then $x =$

a. $\sin^{-1} y$ for $y \in [0, 1]$ only

b. $\cos^{-1} y$

c. π/y

d. $\arcsin y$ for $y \in [-1, 1]$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions & Their Graphs

State the angle addition formula $\sin(A + B) =$

a. $\cos A\cos B - \sin A\sin B$

b. $\sin A\cos B + \cos A\sin B$

c. $\sin A\cos B - \cos A\sin B$

d. $\sin A + \sin B$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Matrices & Systems

State the double-angle formula $\sin 2\theta =$

a. $2\cos^2\theta - 1$

b. $\sin^2\theta - \cos^2\theta$

c. $2\sin\theta\cos\theta$

d. $\cos^2\theta - \sin^2\theta$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Probability & Statistics

Find the reference angle for 210°.

a. 45°

b. 60°

c. 150°

d. 30°

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Convert $\pi/3$ radians to degrees.

a. 45°

b. 60°

c. 90°

d. 30°

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Polynomial Functions

For $y = \sin x$, the x-intercepts occur at:

a. $x = (2n + 1)\pi/2$

b. $x = 2n\pi$

c. $x = n\pi/2$

d. $x = n\pi,\ n \in \mathbb{Z}$

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